# Euiseok Han

hanes1207@kaist.ac.kr hanes1207.github.io linkedin.com/in/hanes1207

## Research Interests

•Uncertainty-Aware Decision Making for Autonomous Systems: Multimodal Perception, State Estimation, Robot Policies, 3D Scene Understanding

## Education

**•Korea Advanced Institute of Science and Technology, Daejeon** *Mar. 2020 – Aug. 2027 (Expected)*
*B.S. in Electrical Engineering and Computer Science (Double Major)* GPA: 4.02/4.3

– *Coursework*: Machine Learning for 3D Data, Kalman Filtering (graduate), Mathematical Physics

**•University of Texas at Austin** *Aug. 2025 – Dec. 2025*
*Exchange Program | University Honors (Fall 2025)* GPA: 4.0/4.0

– *Coursework*: Digital Signal Processing, Classical Electrodynamics, Quantum Physics

## Publications

[1] **SenseFuse: Label-Free Fusion of Image and Shape Encoders for Open-Vocabulary 3D Instance Segmentation** *Under review*
**Euiseok Han**, Tri Ton, Hwanhee Kim, Seungyeon Ryu, Chang D. Yoo
*IEEE Int. Conf. on Robotics and Automation (ICRA) 2027. arXiv:2609.20475*
https://hanes1207.github.io/sensefuse

[2] **Reference-Filter-Driven Transition Probabilities for IMM-Based Satellite Maneuver Detection** *Preprint*
**Euiseok Han**, Sangheon Choi, Seung-Hyun Kong
*arXiv preprint arXiv:2610.11242*
https://hanes1207.github.io/rdimm

## Research Experience

**•Artificial Intelligence and Machine Learning Lab, KAIST** *Jan. 2025 – Feb. 2025, Jan. 2026 – Present*
*Undergraduate Researcher | Advisor: Prof. Chang D. Yoo* Daejeon, KR

– URP: *From Point Cloud to Actions: 3D Structural Keypoint-Conditioned Hierarchical Robot Policy Learning*
– Revealed that frozen open-vocabulary 2D image and 3D shape encoders fail on largely disjoint instances and share only 5.9% of their wrong labels, versus 41.4% for a pair of 2D encoders [1]
– Devised a 2D–3D fusion framework with a closed-form, label-free scene-level weight via sensitivity maximization (3.5 ms), achieving a median gain of +7.1 points and 93% of the oracle-weight gain
– Implemented Gymnasium and joystick teleoperation interfaces for a UR5 robot; trained a diffusion policy on the collected demonstrations

**•Cho Chun Shik Graduate School of Mobility, KAIST** *Mar. 2026 – Present*
*Undergraduate Researcher | Advisor: Prof. Seung-Hyun Kong* Daejeon, KR

– Identified a feedback loop in adaptive-TPM IMMs: statistics computed inside the IMM depend on the TPM through mixing, which sustains false alarms or suppresses detections
– Proposed RD-IMM, driving the transitions from the NIS of a reference coast filter outside the IMM [2]
– Detected 98% of 1 m/s north–south (out-of-plane) burns missed by a fixed-TPM IMM, with few false alarms

**•NC Lab, KAIST** *Mar. 2025 – Aug. 2025*
*Undergraduate Researcher | Advisor: Prof. Junehwa Song* Daejeon, KR

– *Socially-appropriate Use of Noise-making Everyday Appliances in High-density Urban Housing*
– Participated in experiment design and conducted a preliminary experiment using Zigbee modules

**•Control and Intelligent Systems Lab, KAIST** *Jul. 2024 – Aug. 2024*

*Undergraduate Researcher | Advisor: Prof. Heejin Ahn* Daejeon, KR

– Reproduced Constrained Policy Optimization (ICML 2017) baselines in MuJoCo (safe RL)

## Selected Projects

**•LLM-Guided Autonomous Robots** *Mar. 2026 – Jun. 2026*

*EE40078 Introduction to Multi-disciplinary Robotics, School of Electrical Engineering*

– LLM-guided autonomous mobile delivery robot; RTAB-Map localization, path planning, and object pick-up
– Built the RGB-D grasping pipeline with a Kalman-filtered target estimate and the ROS mission orchestrator

**•PintOS-KAIST** *Sep. 2024 – Dec. 2024*

*CS330 Operating Systems, School of Computing*

– Implemented kernel modules for process scheduling, syscall, memory management of an educational x86-64 OS

## Honors and Awards

**•Korea-U.S. Advanced Industries Youth Exchange Scholarship** *Aug. 2025*

$9,000 for advanced-industry exchange study in the U.S.

## Other Experience

**•Republic of Korea Air Force (mandatory military service)** *Apr. 2022 – Jan. 2024*

*Information System Operator, Staff Sergeant, KAMD Operations Center* Osan Air Base, KR

– Operated 24/7 real-time data processing servers for the Korea Air and Missile Defense center

## Technical Skills and Interests

**Programming Languages**: C, C++, Python
**Libraries**: ROS, NumPy, PyTorch, Gymnasium, MuJoCo
**Miscellaneous tools**: MATLAB, LaTeX, Docker
**Languages**: Korean (native), English (TOEFL iBT 5.5/6)
**Extracurricular**: STROKE (Tennis Club)